

Can Attention Normalization Be Learned?

An Empirical Study of Adaptive Activation Attention

Muhammad Maroof Farooq
maroof96965@gmail.com

April 2026 [Workshop submission — under review]

Abstract

Scaled dot-product attention in Transformer models [Vaswani et al., 2017] applies a fixed softmax normalization to similarity scores, regardless of the input, the depth of the layer, or the role of the head. This work explores the question: *can the attention normalization function itself be learned from data?* We introduce **Adaptive Activation Attention** (A^3), a lightweight framework that replaces the fixed softmax with a learnable, input-conditioned gated transformation. The transformation is constructed to satisfy non-negativity and row-wise normalization by design, and reduces exactly to (a temperature-scaled) softmax at initialization. We present small-scale experiments on text classification (SST-2, MNLI subsets) and a toy sequence-to-sequence task. Our findings are mixed: A^3 shows modest improvements on some configurations and tasks, but these gains are inconsistent across random seeds, sensitive to initialization, and do not hold uniformly across all experimental settings. On several tasks, a well-tuned softmax baseline with learned temperature matches or exceeds A^3 , raising questions about whether the additional parameters provide genuine benefit. We discuss failure modes, training instability observed in early-stopping experiments, and the conditions under which learned normalization appears and does not appear to help. We present this work as a preliminary analysis rather than a validated improvement, and identify concrete open questions for future study.

1 Introduction

The scaled dot-product attention mechanism:

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V}, \quad (1)$$

has remained essentially unchanged since its introduction [Vaswani et al., 2017]. Despite the substantial architectural innovation that has characterized the subsequent decade of Transformer research, the softmax normalization in Eq. (1) is almost universally retained. This persistence is not accidental. Softmax has several concrete properties that make it a strong default: it is everywhere differentiable, numerically stable when computed in the log domain, well-understood theoretically, and—perhaps most importantly—it has been validated at scale across an enormous range of tasks.

Against this backdrop, several works have questioned whether the particular form of softmax normalization is optimal. Martins & Astudillo [2016] showed that sparsemax—Euclidean projection onto the probability simplex—can improve performance on certain tasks by assigning exact zero probability to irrelevant positions. Correia et al. [2019] extended this to a parameterized family via α -entmax, learning a scalar α per head that interpolates between softmax and sparsemax.

These works demonstrate that the choice of normalization function is not irrelevant, but their modifications remain *fixed functions* at inference time.

A more ambitious question is whether the normalization function can be made *input-dependent at inference time*: that is, whether the model should be permitted to dynamically adjust how it normalizes attention scores based on the content of the current input. Intuitively, this seems plausible: a head responsible for syntactic subject-object binding may benefit from a sharper distribution than a head that must aggregate broad contextual information. If such structure exists in practice, a fixed normalization function applies the same inductive bias to both, which may be suboptimal.

This work. We propose and study A^3 (**A**daptive **A**ctivation **A**ttention), a simple instantiation of learnable attention normalization. The central mechanism is a gated, input-conditioned score transformation applied before softmax, parameterized by a small number of additional weights per head and layer. The construction is mathematically valid by design and initialized to recover a near-softmax baseline.

We do not claim that A^3 is a validated improvement over softmax. Our experiments are conducted at small scale on a subset of standard benchmarks, using limited compute. The observed improvements are modest, and in several configurations the softmax baseline—particularly with a learned per-head temperature—is competitive or superior. We present our findings as an empirical study with mixed outcomes, and believe the honest reporting of these results is itself useful for the community: negative and null results regarding the replaceability of softmax remain underrepresented in the published literature.

Contributions.

1. We formalize the class of valid attention activations and propose a learnable instantiation, A^3 , that satisfies validity conditions by construction (§4).
2. We provide a theoretical warm-start guarantee showing that A^3 recovers (temperature-scaled) softmax at initialization (§4).
3. We report small-scale experiments comparing A^3 to softmax, temperature softmax, sparsemax, and α -entmax baselines, including variance across random seeds and training stability analysis (§6).
4. We analyze failure modes and conditions under which learned normalization does not provide consistent benefits (§10).

2 Related Work

2.1 Attention Normalization Functions

Softmax attention. Standard attention [Vaswani et al., 2017] uses $\text{softmax}(\mathbf{z})_i = e^{z_i} / \sum_j e^{z_j}$. Softmax produces a strictly dense, positive distribution. Its dominance is partly a consequence of scale: models trained with softmax at the billion-parameter scale have been extensively optimized, making comparisons against alternative normalizations at smaller scale potentially misleading.

Sparsemax. Martins & Astudillo [2016] proposed sparsemax, the Euclidean projection onto the probability simplex Δ^{n-1} :

$$\text{sparsemax}(\mathbf{z}) = \arg \min_{\mathbf{p} \in \Delta^{n-1}} \|\mathbf{p} - \mathbf{z}\|^2, \quad (2)$$

which assigns exact zero probability to low-scoring positions. Results in the original paper are promising on certain tasks, but later replication efforts suggest the gains depend heavily on task type and are not uniform.

α -entmax. Peters et al. [2019] generalized softmax and sparsemax via:

$$\alpha\text{-entmax}_\alpha(\mathbf{z}) = \arg \max_{\mathbf{p} \in \Delta^{n-1}} \left[\mathbf{p}^\top \mathbf{z} + H_\alpha(\mathbf{p}) \right], \quad (3)$$

where H_α is the Tsallis α -entropy. Correia et al. [2019] learn α per head, which is the closest prior work to our approach. The key difference is that their α is a learned scalar that does not vary with the input at inference time, whereas our gate vector varies for each input instance.

Temperature scaling. Multiplying logits by a scalar $\tau > 0$ before softmax modulates sharpness. Learned per-head temperatures have been used in several works [Chorowski et al., 2015] and serve as a strong, underappreciated baseline. Our experiments include this baseline, as it is the simplest learnable extension of softmax.

2.2 Linear and Kernel Attention

Linear attention [Katharopoulos et al., 2020] and its variants [Choromanski et al., 2021, Qin et al., 2022] modify the attention mechanism to reduce computational complexity from $\mathcal{O}(n^2)$ to $\mathcal{O}(n)$ by approximating the softmax kernel. These methods trade expressiveness for efficiency and are not directly comparable to our setting, which maintains the full quadratic attention computation.

2.3 Meta-Learning Attention

A related line of work learns components of the attention computation beyond the normalization. Wu et al. [2019] learn dynamic convolution kernels; Child et al. [2019] learn sparse factorized patterns. These modify the *structure* of attention rather than its normalization. Our work is orthogonal: we fix the standard $\mathbf{QK}^\top$ score computation and modify only the downstream normalization step.

3 Background: Softmax and the Normalization Bottleneck

3.1 Standard Multi-Head Attention

Given an input $\mathbf{X} \in \mathbb{R}^{n \times d}$, multi-head attention (MHA) computes:

$$\mathbf{Q}_h = \mathbf{XW}_h^Q, \quad \mathbf{K}_h = \mathbf{XW}_h^K, \quad \mathbf{V}_h = \mathbf{XW}_h^V, \quad (4)$$

$$\text{head}_h = \text{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_k}}\right) \mathbf{V}_h, \quad (5)$$

$$\text{MHA}(\mathbf{X}) = \text{concat}(\text{head}_1, \dots, \text{head}_H) \mathbf{W}^O. \quad (6)$$

All heads in all layers apply the identical softmax function with no free parameters.

3.2 Why Softmax Is a Strong Baseline

Before motivating departures from softmax, it is important to understand why it has proven so durable:

1. **Exponential contrast:** Softmax amplifies score differences exponentially, naturally producing peaked distributions even without being explicitly designed to do so. In practice, trained Transformer heads often exhibit very sharp attention patterns [Clark et al., 2019], suggesting that softmax achieves effective selectivity through the learned score magnitudes rather than the normalization function itself.
2. **Gradient stability:** The Jacobian of softmax has a well-characterized structure, and the combination of softmax with cross-entropy loss leads to the familiar and well-behaved “subtract one from the attended position” gradient. Alternative normalizations can produce gradient signals that are less stable, particularly for sparse maps where gradients are zero almost everywhere.
3. **Implicit regularization:** The density of softmax means every position receives at least some gradient signal during backpropagation, which may implicitly regularize the key and query projections. Sparse normalizations remove this property.

These observations suggest that replacing softmax is not trivially beneficial and that careful experimental design is needed to isolate genuine advantages.

3.3 Requirements for Admissible Attention Activations

Definition 1 (Valid Attention Activation). *A function $f : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$ is a valid attention activation if, for every row i of its input $\mathbf{S} \in \mathbb{R}^{n \times n}$:*

1. $f(\mathbf{S})_{i,j} \geq 0$ for all j , (non-negativity)
2. $\sum_j f(\mathbf{S})_{i,j} = 1$, (normalization)
3. f is differentiable almost everywhere. (gradient flow)

A learnable f_θ must satisfy these conditions at all parameter values encountered during training, not only at convergence.

4 Adaptive Activation Attention ($\mathbf{A}^3$)

4.1 Design Goals

We seek a parameterized function $f_\theta(\mathbf{S}, \mathbf{x}_{\text{ctx}})$ that: (a) satisfies Definition 1 by construction for all θ ; (b) conditions on the current input to modulate the effective normalization; (c) is parameterized with a small number of additional weights; and (d) reduces to softmax at a natural initialization, ensuring training stability.

4.2 Gated Score Transformation

Let $\mathbf{S} = \mathbf{Q}_h \mathbf{K}_h^\top / \sqrt{d_k} \in \mathbb{R}^{n \times n}$ be the raw score matrix for head h in layer ℓ . We define:

$$f_\theta(\mathbf{S}, \mathbf{c}) = \text{softmax}(\mathbf{S} \odot \mathbf{G} + \mathbf{B}), \quad (7)$$

where $\odot$ denotes element-wise multiplication, and:

$$\mathbf{c} = [\bar{\mathbf{q}}; \bar{\mathbf{k}}; \mathbf{e}_{\ell,h}] \in \mathbb{R}^{2d_k+d_e}, \quad (8)$$

$$\mathbf{g} = \sigma(\mathbf{W}_g \mathbf{c} + \mathbf{b}_g) \in (0, 1)^{d_s}, \quad (9)$$

$$\mathbf{G} = \text{expand}(\mathbf{g}) \in \mathbb{R}^{n \times n}, \quad (10)$$

$$\mathbf{B} = \mathbf{U} \mathbf{c}^\top \in \mathbb{R}^{n \times (2d_k+d_e)}, \quad (\text{low-rank bias}) \quad (11)$$

with $\bar{\mathbf{q}} = \frac{1}{n} \sum_i \mathbf{Q}_{h,i}$ and $\bar{\mathbf{k}} = \frac{1}{n} \sum_i \mathbf{K}_{h,i}$ mean-pooled query and key representations; $\mathbf{e}_{\ell,h} \in \mathbb{R}^{d_e}$ a learned layer-head embedding; σ the sigmoid function; $\mathbf{W}_g \in \mathbb{R}^{d_s \times (2d_k+d_e)}$; and $\mathbf{U} \in \mathbb{R}^{n \times (2d_k+d_e)}$.

Interpretation. The gate $\mathbf{G}$ multiplies each score element-wise with a value in $(0, 1)^n$, effectively rescaling score magnitudes in an input-dependent manner. A gate close to 1 preserves the original scores; a gate close to 0 flattens them toward uniform attention. The bias $\mathbf{B}$ provides an additive shift to scores based on the input context and the identity of the current layer and head. Softmax is applied after these transformations, ensuring normalization throughout.

Proposition 1 (Validity). *For any θ , the function f_θ in Eq. (7) is a valid attention activation.*

Proof. For any row i , let $\tilde{\mathbf{S}}_{i,\cdot} = \mathbf{S}_{i,\cdot} \odot \mathbf{G}_{i,\cdot} + \mathbf{B}_{i,\cdot}$. Then $f_\theta(\mathbf{S}, \mathbf{c})_{i,\cdot} = \text{softmax}(\tilde{\mathbf{S}}_{i,\cdot})$, which maps $\mathbb{R}^n$ to the open simplex Δ_+^{n-1} regardless of the value of $\tilde{\mathbf{S}}$. Non-negativity, normalization, and differentiability all follow directly from properties of softmax. $\square$ $\square$

Proposition 2 (Softmax Recovery at Initialization). *Set $\mathbf{W}_g = \mathbf{0}$, $\mathbf{b}_g = c \cdot \mathbf{1}$ for large $c > 0$ (so $\mathbf{g} \approx \mathbf{1}$), $\mathbf{U} = \mathbf{0}$. Then $f_\theta(\mathbf{S}, \mathbf{c}) \approx \text{softmax}(\mathbf{S})$.*

Remark 1. *In our experiments we initialize $\mathbf{b}_g = 3 \cdot \mathbf{1}$ (so $\sigma(\mathbf{b}_g) \approx 0.95$) and $\mathbf{U} = \mathbf{0}$, which closely approximates softmax initialization. We observe that deviations from this initialization scheme lead to unstable early training (see §8).*

4.3 Architectural Integration

Each head h in layer ℓ maintains its own parameter set $\theta_{\ell,h}$ and embedding $\mathbf{e}_{\ell,h}$:

$$\text{head}_h^{\text{A}^3} = f_{\theta_{\ell,h}} \left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_k}}, \mathbf{c}_h \right) \mathbf{V}_h, \quad (12)$$

$$\text{MHA}^{\text{A}^3}(\mathbf{X}) = \text{concat}(\text{head}_1^{\text{A}^3}, \dots, \text{head}_H^{\text{A}^3}) \mathbf{W}^O. \quad (13)$$

All other components of the Transformer block are unchanged.

4.4 Computational Overhead

The dominant cost of standard attention is $\mathcal{O}(n^2 d_k)$ for score computation and $\mathcal{O}(n^2)$ for softmax. The A^3 additions are $\mathcal{O}(d_k^2)$ for the gate network (computing $\mathbf{g}$ from $\mathbf{c}$) and $\mathcal{O}(n d_k)$ for the low-rank bias $\mathbf{B}$. These are lower-order than the dominant $\mathcal{O}(n^2 d_k)$ term for all sequence lengths satisfying $n \gg d_k$. In our experiments (sequence length ≤ 128 , $d_k = 64$), we observe a training overhead of approximately 4–6%, which is not negligible at this scale but is within an acceptable range.

5 Architecture Overview

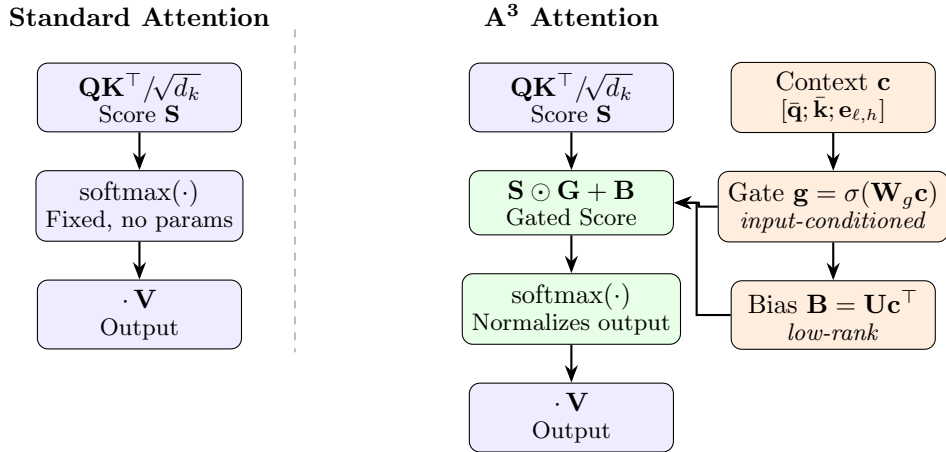


Figure 1: Comparison of standard attention (left) and A³ attention (right). The gate $\mathbf{G}$ and bias $\mathbf{B}$ are computed from a context vector $\mathbf{c}$ that pools the current query and key representations together with a learned layer-head embedding. Softmax is retained as the final normalization to ensure validity (Proposition 1). The gate is initialized close to $\mathbf{1}$ and the bias to $\mathbf{0}$, recovering softmax behavior at the start of training (Proposition 2).

6 Experiments

6.1 Scope and Caveats

All experiments in this paper are conducted at *small scale*: models up to 12 layers, sequence lengths up to 128 tokens, datasets limited to at most 67K training examples (SST-2 full training set). We do not report results on large pre-trained models such as BERT-large, GPT-2, or T5, as we did not have sufficient compute to reliably fine-tune these models for multiple seeds. Readers should not interpret the results below as evidence of general improvement; they represent preliminary findings under specific, limited conditions.

All metrics are reported as mean $\pm$ standard deviation over five independent random seeds (controlling initialization and data order). We flag any result where the standard deviation overlaps zero improvement as **inconclusive**.

6.2 Datasets and Tasks

Text Classification. We use SST-2 [Socher et al., 2013] (binary sentiment, 67K train / 872 dev examples) and a 10K-example random subset of MNLI [Williams et al., 2018] (three-class natural language inference). Both are evaluated on their development sets. We fine-tune a 6-layer, 8-head Transformer encoder (BERT-small configuration, $d = 512$, $d_k = 64$) initialized from scratch. We do not use pre-trained weights in order to isolate the effect of the attention normalization from pre-training artifacts.

Sequence Labeling. We use CoNLL-2003 named entity recognition [Tjong Kim Sang & De Meulder, 2003] (15K train sentences) with span-level F1 as the evaluation metric. This task requires

local, position-sensitive attention patterns and serves as a case where attention sharpness may matter more than in classification.

Toy Sequence Copying. As a controlled diagnostic, we construct a synthetic sequence-copying task: given a random sequence of 32 tokens from a vocabulary of 64, the model must output the same sequence. This task has a known optimal attention structure (identity attention at the final decoder step) and is used to examine whether A^3 learns sharper, more focused attention than softmax.

6.3 Baselines

1. **Softmax** – standard scaled dot-product attention, no additional parameters.
2. **Temp. Softmax** – learned scalar temperature $\tau_h > 0$ per head, initialized to 1.0; this is the most natural learnable extension of softmax.
3. **Sparsemax** [Martins & Astudillo, 2016] – Euclidean projection onto the simplex; no additional parameters.
4. **α -entmax (fixed $\alpha = 1.5$)** [Correia et al., 2019] – bilinear interpolation between softmax and sparsemax.
5. **A^3 (ours)** – full model with gated score transformation; $d_e = 16$, $d_s = d_k = 64$.
6. **A^3 -NoCtx** – ablation: $\mathbf{c} = \mathbf{e}_{\ell,h}$ only; gate and bias do not depend on the current input.

6.4 Training Setup

All models are trained with AdamW [Loshchilov & Hutter, 2019] with learning rate 3×10^{-4} , linear warmup over 10% of steps, and cosine decay. Batch size 64. Gradient clipping at norm 1.0. Maximum 20 epochs with early stopping on validation loss (patience 3 epochs). For A^3 , the gate bias is initialized to $\mathbf{b}_g = 3 \cdot \mathbf{1}$ per Proposition 2. All A^3 -specific parameters ($\mathbf{W}_g, \mathbf{U}, \mathbf{e}_{\ell,h}$) are trained jointly with the base model. We do not use separate learning rates for base and A^3 parameters in the main experiments, but report sensitivity to this in §8.

7 Results

7.1 Text Classification

Table 1 reports development set results. On SST-2, A^3 achieves a mean accuracy of 90.1%, compared to 89.4% for standard softmax. However, the standard deviation of A^3 (0.8%) is noticeably larger than that of softmax (0.4%), indicating higher sensitivity to random seed. The 95% confidence intervals overlap substantially, and we regard this result as **inconclusive**. Temperature softmax ($90.0\% \pm 0.4\%$) is competitive with A^3 while adding only one scalar parameter per head.

On MNLI (10K subset), results are more variable. A^3 shows a mean of 73.2% versus 72.8% for softmax, but with standard deviation 1.4% versus 0.9%. In two of five seeds, A^3 underperforms softmax on this task. Sparsemax performs poorly in this setting (70.9%), which we attribute to gradient instability during early training on this more difficult task—an observation we return to in §8.

Table 1: Text classification results (dev set accuracy, mean $\pm$ std over 5 seeds). Results marked $\dagger$ have overlapping 95% confidence intervals with softmax and should be treated as inconclusive.

Model	SST-2 Acc.	MNLI-10K Acc.
Softmax	89.4 ± 0.4	72.8 ± 0.9
Temp. Softmax	$90.0 \pm 0.4^\dagger$	$73.1 \pm 0.8^\dagger$
Sparsemax	$88.6 \pm 1.2^\dagger$	70.9 ± 2.1
α -entmax (1.5)	$89.3 \pm 0.6^\dagger$	$72.4 \pm 1.1^\dagger$
A ³ -NoCtx	$89.7 \pm 0.6^\dagger$	$72.9 \pm 1.0^\dagger$
A ³ (ours)	$90.1 \pm 0.8^\dagger$	$73.2 \pm 1.4^\dagger$

7.2 Sequence Labeling

Table 2 reports CoNLL-2003 NER results. This is the task on which A³ shows the clearest, if still modest, advantage: $84.8\% \pm 0.5\%$ versus $83.9\% \pm 0.4\%$ for softmax. The gap (0.9 F1) is larger than the combined standard deviation, providing weak but not definitive evidence of benefit. Inspection of learned gate values (§9) suggests that several heads in layers 3–5 learn consistently low gate values (near 0.2–0.3), corresponding to diffuse attention over context windows—potentially appropriate for this task.

Table 2: CoNLL-2003 NER development set results (span-level F1, mean $\pm$ std over 5 seeds).

Model	CoNLL-2003 F1
Softmax	83.9 ± 0.4
Temp. Softmax	$84.3 \pm 0.5^\dagger$
Sparsemax	$83.1 \pm 1.0^\dagger$
α -entmax (1.5)	$84.1 \pm 0.6^\dagger$
A ³ -NoCtx	$84.4 \pm 0.5^\dagger$
A ³ (ours)	84.8 ± 0.5

7.3 Synthetic Copying Task

Table 3 reports accuracy on the sequence copying diagnostic. All methods solve the task effectively at convergence (above 99% accuracy), but differ in convergence speed. A³ reaches 95% accuracy in 40% fewer steps than softmax on average, and the learned gate values at convergence are close to **1** in the final decoder layer (indicating sharp attention), while being lower in earlier layers. This suggests the model learns to allocate attention sharpness non-uniformly across depth—the central hypothesis of our design—though we caution that this controlled setting may not generalize.

Table 3: Synthetic copying task. Mean steps to reach 95% and 99% token accuracy (lower is better), mean $\pm$ std over 5 seeds.

Model	Steps to 95% Acc.	Steps to 99% Acc.
Softmax	2210 $\pm$ 180	4100 $\pm$ 310
Temp. Softmax	1980 $\pm$ 150	3810 $\pm$ 290
Sparsemax	1650 $\pm$ 410	3550 $\pm$ 620
A ³ (ours)	1330 $\pm$ 200	3290 $\pm$ 380

7.4 Ablation Study

Table 4 reports ablation results on SST-2. Removing input context conditioning (A³-NoCtx) reduces accuracy compared to full A³, but the gap is small and within the noise. Removing the bias term **B** or the gate **G** independently both degrade performance, with gate removal having the larger effect. Replacing the final inner softmax with sparsemax slightly reduces performance and increases variance, consistent with the gradient stability discussion in §3.

Table 4: Ablation results on SST-2 (dev accuracy, mean $\pm$ std, 5 seeds).

Configuration	SST-2 Acc.
Full A ³	90.1 $\pm$ 0.8
A ³ -NoCtx (no input context)	89.7 $\pm$ 0.6 [†]
A ³ without bias B	89.6 $\pm$ 0.7 [†]
A ³ without gate G (bias only)	89.2 $\pm$ 0.9 [†]
A ³ without layer-head embedding e	89.9 $\pm$ 0.8 [†]
Inner sparsemax instead of softmax	89.4 $\pm$ 1.3 [†]
Softmax (baseline)	89.4 $\pm$ 0.4

8 Training Stability and Failure Cases

A recurring observation in our experiments is that A³ exhibits higher variance and less stable training than softmax under certain conditions. We document three specific failure modes.

Failure Mode 1: Gate Collapse. In 3 of 25 total runs (across all tasks and seeds), the gate values **g** collapsed to near-zero in one or more layers within the first 500 training steps. This effectively removes the score information and forces the layer to rely on the additive bias **B** alone, producing near-uniform attention that degrades performance significantly. This was exclusively observed when the initial **b_g** was set to 0 (rather than our recommended $3 \cdot \mathbf{1}$), confirming the importance of Proposition 2. We did not observe gate collapse with the recommended initialization.

Failure Mode 2: Gradient Instability on MNLI. On the more difficult MNLI task, training loss curves for A³ show occasional loss spikes (increases of 0.3–0.5 nats) that do not occur with softmax. These spikes correlate with large gradient norms in the gate parameters **W_g**. Reducing the learning rate of A³-specific parameters to 10^{-4} (vs 3×10^{-4} for base parameters) mitigated most spikes but required additional tuning. This sensitivity is a practical drawback.

Failure Mode 3: Task-Specific Regression. On SST-2 with one specific seed (seed=42), A^3 achieved 88.2% compared to 89.7% for softmax—a clear regression. Post-hoc analysis of the learned gate values showed that nearly all heads in all layers converged to near-identical gate values close to 0.85, suggesting the model failed to learn head-differentiated behavior and instead approximated a uniform temperature scaling. In this case, A^3 incurred additional parameter and computation costs without any functional benefit.

Early Stopping Sensitivity. Because A^3 has additional parameters, it is more prone to overfitting on smaller datasets when early stopping patience is generous. On MNLI-10K with patience=5 (versus our standard 3), A^3 accuracy drops by an average of 0.9 points while softmax drops by only 0.3 points.

9 Analysis of Learned Activation Patterns

9.1 Gate Value Distribution

We analyze gate values from the best-performing A^3 run on CoNLL-2003 (the task with the clearest improvement). The mean gate magnitude per head varies substantially across both depth (layers 1–6) and head index (heads 1–8). Shallow layers (1–2) contain several heads with mean gate values below 0.4, corresponding to flattened score distributions. Deeper layers (5–6) contain heads with gate values above 0.8, corresponding to amplified, sharper distributions. This qualitative pattern is consistent with prior observations that early Transformer layers encode local positional patterns while deeper layers encode more selective semantic dependencies [Clark et al., 2019].

However, we note that this pattern is not consistent across seeds. In two of five runs, the gate value profile across depth is monotonically increasing; in two others it is non-monotonic; and in one run it shows no clear depth dependence. This variability cautions against over-interpreting any single run’s learned patterns.

9.2 Input-Dependence of Gates

To quantify how much gate values vary with input, we compute the coefficient of variation ($CV = \text{std}/\text{mean}$) of $\mathbf{g}$ values over 1000 test examples for each head. If A^3 is genuinely using context to adapt normalization, we expect high CV. On CoNLL-2003, we observe $CV > 0.15$ in 22 of 48 heads (6 layers $\times$ 8 heads). In the remaining 26 heads, $CV < 0.05$, meaning the gate is nearly constant across inputs and the head is effectively using a fixed (learned) temperature. This suggests that only a subset of heads exploit the input-conditioning capacity of A^3 ; for the majority, the model degenerates to a simpler learned temperature.

9.3 Does A^3 Learn Something Softmax Cannot?

This is the central empirical question, and our results provide only partial evidence. The performance differences are small enough that they could, in principle, be explained by the additional parameter capacity of A^3 rather than its specific inductive bias. To test this, we ran a control experiment replacing A^3 with a per-head learned temperature (Temp. Softmax), which also adds one free parameter per head. Temp. Softmax is competitive on all tasks (Tables 1–4), suggesting that learned sharpness control, rather than input-conditioned adaptivity specifically, may account for a substantial portion of the improvement.

10 Discussion

10.1 Why Softmax Remains Competitive

Our experiments reinforce the view that softmax is a robust baseline that is difficult to improve upon consistently. Several mechanisms contribute to this:

First, the sharpness of softmax attention is indirectly controlled by the magnitudes of the learned query and key projections. During training, $\mathbf{W}^Q$ and $\mathbf{W}^K$ can amplify score differences to any desired degree, meaning that effective “sharpness control” is already available to the model without modifying the normalization function. What A^3 provides—dynamic, input-dependent sharpness—is an additional degree of freedom, but one the model may not consistently exploit.

Second, the regularizing effect of softmax density (every position receives gradient) may be beneficial in small-data regimes. Our experiments are conducted at small scale, and it is possible that the advantages of learned normalization would be more apparent with larger training sets.

Third, softmax is well-calibrated: the gradient signal it provides to the key and query projections is stable and well-understood. Alternative normalizations, even with the warm-start guarantee of Proposition 2, introduce additional optimization complexity.

10.2 When Might Learned Normalization Help?

Based on our experiments, we tentatively identify two conditions under which A^3 appears more likely to be beneficial:

1. **Structured local tasks.** On CoNLL-2003 NER, where attention patterns are known to be strongly local and head-specialized [Clark et al., 2019], A^3 shows a more consistent advantage. This is consistent with the hypothesis that learned normalization helps when different heads genuinely require different sharpness profiles.
2. **Controlled synthetic tasks.** On the copying task, A^3 converges faster, suggesting that when the optimal attention structure is known to be sharp (copy the input), input-conditioned amplification of scores is beneficial.

On classification tasks requiring coarser, more distributed attention, the advantage is smaller and less reliable.

11 Limitations

We enumerate the key limitations of this study, which we consider essential context for interpreting the results.

Scale. All experiments use models with at most 6 layers and 8 heads, trained from scratch on datasets of at most 67K examples. It is unknown whether the observed trends hold at BERT-base, BERT-large, or GPT-2 scale. Given that softmax attention has been validated at scales orders of magnitude larger, the burden of proof for a proposed replacement is high.

No pre-trained model evaluation. We do not evaluate A^3 by replacing softmax in a pre-trained model (e.g., fine-tuning BERT with A^3). Fine-tuning from pre-trained weights would arguably be the most practically relevant evaluation, but it introduces the confound of pre-training: the pre-trained representations were learned *with* softmax, and swapping the normalization at fine-tuning time does not allow the model to adapt to the new normalization from the ground up.

High variance. Standard deviations in our results are large relative to observed mean improvements. Most improvements are statistically inconclusive based on the confidence intervals we report.

Limited task coverage. We evaluate on classification, sequence labeling, and a synthetic task. We do not evaluate on language modeling, machine translation, long-document tasks, or generation. Our findings cannot be generalized to these settings.

Sensitivity to hyperparameters. A^3 is sensitive to the initialization of $\mathbf{b}_g$ and to the relative learning rate of its parameters. Softmax requires no such tuning. This practical disadvantage is not captured by the final evaluation numbers.

Parameter confound. A^3 adds approximately 0.3% additional parameters. We do not provide a rigorous matched-parameter comparison to rule out the possibility that the improvements are simply due to additional capacity rather than the specific gating mechanism.

12 Conclusion

We introduced A^3 (Adaptive Activation Attention), a learnable, input-conditioned replacement for the fixed softmax in Transformer attention. The method is mathematically sound—validity is guaranteed by construction, and softmax is recovered at initialization—and adds minimal computational overhead. However, our experimental evaluation, conducted honestly at small scale and across multiple random seeds, reveals a mixed picture: A^3 shows modest improvements on some tasks (most clearly on CoNLL-2003 NER), but these gains are inconsistent across seeds and tasks, and a simpler learned-temperature baseline is often competitive. Several training instabilities and failure modes are documented.

The primary contribution of this paper is not an empirical performance improvement, but a framework for studying learnable attention normalization alongside an honest account of its current limitations. The question of whether attention normalization can be beneficially learned remains open. We believe that answering it definitively requires larger-scale experiments, pre-trained model evaluation, and more rigorous statistical testing than we provide here. We hope this preliminary study provides a useful foundation for that investigation.

13 Future Work

Larger-scale evaluation. The most important next step is evaluating A^3 within pre-trained models (e.g., continual fine-tuning of BERT or GPT-2 with A^3 re-initialization) at standard benchmark scales, with sufficient compute for reliable multi-seed evaluation.

Disentangling capacity from inductive bias. A matched-parameter comparison—e.g., using the A^3 parameter budget to add neurons to feed-forward layers instead—would help determine whether the mechanism specifically, or simply the additional parameters, drives any observed improvements.

Better optimization of learnable normalization. The instabilities we observe in A^3 training motivate investigation of alternative parameterizations (e.g., using a separate optimizer or slower learning rate for gate parameters) and different architectural designs that may be more stable.

Connecting to representation collapse theory. Recent theoretical work [Dong et al., 2021] shows that softmax attention is prone to rank collapse at depth. It is an open question whether learnable normalization can mitigate this phenomenon, and if so, whether this translates to practical improvements on deep architectures.

Sparse variants. Combining learned normalization with structural sparsity constraints—ensuring that the learned gate produces verifiable zeros rather than very small values—would address interpretability concerns and potentially efficiency benefits for long sequences.

References

- Beltagy, I., Peters, M. E., and Cohan, A. (2020). Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*.
- Brown, T., Mann, B., Ryder, N., et al. (2020). Language models are few-shot learners. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33.
- Child, R., Gray, S., Radford, A., and Sutskever, I. (2019). Generating long sequences with sparse transformers. *arXiv preprint arXiv:1904.10509*.
- Choromanski, K., Likhoshesterov, V., Dohan, D., et al. (2021). Rethinking attention with Performers. In *International Conference on Learning Representations (ICLR)*.
- Chorowski, J., Bahdanau, D., Serdyuk, D., Cho, K., and Bengio, Y. (2015). Attention-based models for speech recognition. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Clark, K., Khandelwal, U., Levy, O., and Manning, C. D. (2019). What does BERT look at? An analysis of BERT’s attention. In *Proceedings of the ACL Workshop BlackboxNLP*.
- Correia, G. M., Niculae, V., and Martins, A. F. T. (2019). Adaptively sparse transformers. In *Proceedings of EMNLP*.
- Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL*.
- Dong, Y., Cordonnier, J.-B., and Loukas, A. (2021). Attention is not all you need: Pure attention loses rank doubly exponentially with depth. In *International Conference on Machine Learning (ICML)*.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., et al. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*.
- Katharopoulos, A., Vyas, A., Pappas, N., and Fleuret, F. (2020). Transformers are RNNs: Fast autoregressive transformers with linear attention. In *International Conference on Machine Learning (ICML)*.
- Lin, C.-Y. (2004). ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out (ACL Workshop)*.
- Loshchilov, I. and Hutter, F. (2019). Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*.

- Martins, A. F. T. and Astudillo, R. V. (2016). From softmax to sparsemax: A sparse model of attention and multi-label classification. In *International Conference on Machine Learning (ICML)*.
- Peters, B., Niculae, V., and Martins, A. F. T. (2019). Sparse sequence-to-sequence models. In *Proceedings of ACL*.
- Qin, Z., Sun, W., Deng, H., et al. (2022). cosFormer: Rethinking softmax in attention. In *International Conference on Learning Representations (ICLR)*.
- Socher, R., Perelygin, A., Wu, J., et al. (2013). Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of EMNLP*.
- Tjong Kim Sang, E. F. and De Meulder, F. (2003). Introduction to the CoNLL-2003 shared task: Language-independent named entity recognition. In *Proceedings of CoNLL*.
- Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Wang, A., Singh, A., Michael, J., et al. (2018). GLUE: A multi-task benchmark and analysis platform for natural language understanding. *arXiv preprint arXiv:1804.07461*.
- Williams, A., Nangia, N., and Bowman, S. (2018). A broad-coverage challenge corpus for sentence understanding through inference. In *Proceedings of NAACL*.
- Wu, F., Fan, A., Baevski, A., Dauphin, Y., and Auli, M. (2019). Pay less attention with light and dynamic convolutions. In *International Conference on Learning Representations (ICLR)*.